

UBT308: FOOD SCIENCE AND NUTRITION

L	T	P	Cr
3	0	0	3.0

Course Objective: To impart knowledge about the various areas related to food science as a discipline. To develop an understanding of food composition, food quality, analysis and food laws.

Syllabus

Food composition: Introduction to food science as a discipline, Constituents of food (macronutrients and micro-nutrients), functional properties, food sources and deficiency diseases. Other food constituents: Bioactive compounds, anti-nutritional factors, pigments.

Food and nutrition: Malnutrition, recommended dietary allowances (RDAs) for various age groups, the concept of a balanced diet, diet planning. Special dietary needs of people with different medical conditions. Food groups and their classification.

Food microbiology: Food spoilage by micro-organisms: classification of foods on the basis of spoilage, factors affecting the growth of microorganisms in food, chemical changes caused by microorganisms. Foodborne illness (infection and intoxication), pathogenic micro-organisms in food. Desirable relationship of microorganisms with food (fermentation, probiotics).

Food quality and laws and Standards: Food quality & analysis: Quality factors in food physical, chemical and microbiological factors of quality. Sensory evaluation of foods. Sample and sample preparation in foods, proximate analysis of foods. Food laws: voluntary and mandatory food laws in India.

Impetus in Food Industry: New Product Development, strategies, planning for marketing, Process designing of food. Foods types available in the market and need of their innovation.

Course Learning Objectives (CLO)

The students will be able to:

1. Explain importance of different types of nutrients in balanced diet and diet planning.
2. Differentiate between different nutrient components in food and their role in processing and consumption.
3. Correlate basic food microbiology with food safety laws and standards.
4. Determine food quality by food analysis as per food laws and their importance in food industry.
5. Apply traditional methods for food preservation in developing a new food product.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Vaclavik, Vickie, Christian, Elizabeth W - Essentials of Food Science 4th Ed. (2014).
2. Geoffrey Campbell-Platt - Food Science and Technology, Wiley-Blackwell Publisher. 2nd Ed. (2017).
3. Sunetra Roday- Food Science and Nutrition, 2nd Edition, Publisher-Oxford (2012).
4. B. Srilakshmi Food Science, 5th ed. New Age Publishers (2010).
5. L.H. Meyer Food Chemistry, New Age Publishers (2004).
6. Frazier C and Westhoff, C. Food Microbiology, TMH, New Delhi. 5th Ed (2017).

Reference Books

1. S. Suzanne Nielsen-Food Analysis: Food Science Texts Series, Springer; 4th Edition (2010).
2. Avantina Sharma Textbook of Food Science & Technology (Vol - I & II), International Book Distributing Company, 2nd ed. (2010).
3. Fennema's Food Chemistry, 5th Edition (Food Science and Technology, CRC Press; 4 edition (2017).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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